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Hierarchical Privacy-Hardened Federated Defense (HPH-FedDef): A Hybrid Edge-Cloud Federated Framework for Privacy-Preserving Intrusion & Malware Detection in Cyber-Physical Systems

Abstract- The coordinated nature of the cyber-physical attacks and the state of the art of malware in the emerging modern CPS requires the design of collaborative and privacy-aware frameworks of intrusion detection. We present a novel algorithm: Hierarchical Privacy-Hardened Federated Defense (HPH-FedDef) that combines hierarchical device-to-edge-to-cloud federated learning topology, adaptive differential privacy noise scheduling, encrypted update similarity screening and reputation-weighted robust aggregation to provide resilient detection in distributed locations in CPSs. Each Local Device computes lightweight temporal autoencoder anomalies and feature updates for peers to perturb differentially, encrypt, and send for secure aggregation, while each Edge Node stores a tamper-evident ledger to audit and weight-client contributions. It provides further aggregation of consensus with trimmed-mean consensus for global models and releases these signatures in a manner that preserves user privacy at the cloud tier. HPH-FedDef improved upon traditional federated averaging by 9.4 % in average detection F1-score, with 96.7 % under cross CPS datasets and malware collections and non-IID distributions within 15 % malicious client poisoning scenarios, while also ensuring less than 2.3 % membership-inference success (compared to 9.1 %). Fixing the latency overheads for secure aggregation to be less than 12% of baseline makes the approach practicable for near-real-time CPS deployment. Such findings show that HPH-FedDef enables practical and secure joint attack and malware detection in distributed CPS at the proof-of-concept level, filling important gaps in the federated CPS security research landscape.

Keywords- Hierarchical Federated Learning, Privacy-Preserving Intrusion Detection, Cyber-Physical Systems Security, Differential Privacy, Robust Aggregation Techniques

I Introduction

On the other hand, modern cyber-physical systems today behave like well-coordinated, interdependent digital-physical infrastructures which transport sensitive operational data and accommodate diverse devices working under the above critical latency constraints. Coordinated cyber and physical threats targeting these environments further amplify the potential for cyber intrusions and malware to exploit communication gaps, resource limitations and restrictive data-sharing policies. Existing studies highlight the limitations of centralized detection architectures, as they open the systems to confidentiality risks and bottlenecks when device-level data

cannot be transmitted due to regulatory or bandwidth-related constraints [1], [2], [4]. Research additionally emphasizes that sophisticated attackers employment obfuscation methods and delayed household time manipulation strategies, making distal discovery from nodes with varying processing capacities more difficult. With the expansion of CPS infrastructures toward the more autonomous and inter-connected Industry 5.0 environments, there is an urgent need for decentralized defenses that are privacy-preserving and that can provide a long term resilience against adversarial conditions [8], [19], [21]. Federated learning has proven to be a promising direction for decentralized intrusion detection used to guide platform users on the decoupling of model refinement in a non-disclosure way when it comes to raw device telemetry [9]. Overcoming these hurdles is a major step forward, but deployment in hostile CPS may be years away due to a host of unaddressed challenges. The static differential privacy setting typically comes either at the cost of model accuracy degradation or violently breaks the stable differential privacy bound with dynamic data dynamics [1], [12], [17]. Therefore, edge-centric aggregation increases bandwidth and communications efficiency but cannot detect malicious updates (or poisoned gradients) that can subvert global learning [3], [31], [39]. On the other hand, while homomorphic encryption and secure aggregation secure model exchanges, the induced overhead complicates matching real-time responsiveness while also reducing insight into insider manipulation efforts [14,15,27]. In addition to these, CPS heterogeneity, multiple temporal patterns of anomalies, and non-IID (Independent and identically distributed) distributions also worsen the stability of global learning behavior resulting in unresolved trade-off between privacy-preservation and reliable detection performance in distributed environments [7], [13], [30].

To mitigate some of this tension, hybrid edge–cloud strategies have been suggested that enable scale through cross multi-tier infrastructures albeit at the cost of introducing limited communication [24], [33], [34]. However aggregation nodes still have blind spots, where clients can gain a false sense of trust and cause the model parameters to converge to a poisoned one. While reputation-based scoring, behavioral filtering, and strong aggregation are shown to contain poisoning [10], [16], [26], they have been applied within a hierarchical CPS deployment. Furthermore, the detection of malware within federated settings is more challenging due to the existence of polymorphism methods that change the appearance of malware, encrypted payload transport, and behavior tailored based on target environments, which ultimately yield a lack of reproducibility of detection across heterogeneous nodes [22], [23], [29]. This shortcoming highlights the lack of a unified framework that can address at the same time some: (i) intrusion indicators; (ii) malware signatures; (iii) privacy constraints; and (iv) adversarial poisoning in multi-layered CPS architecture. As CPS

infrastructures become more heterogeneous, new operational constraints arise, such as hard timing constraints, lightweight model requirements, and the need to provide verified integrity checks at hierarchical aggregation layers. Targeted poisoning attacks regularly break performance guarantees for traditional federated averaging, leading to increasing calls for privacy-preserving mechanisms with adaptive noise infusion [5], [20], [38], secure similarity screening [18], and transparent auditability [36]. The CPS environment needs to synergistically combine multi-level defense mechanisms, including (i) privacy hardening in local environments, (ii) mutual trust assessment in mid-level environments and (iii) resilient model coordination in the environment between the entities. The above context drives the design of a Hierarchical Privacy-Hardened Federated Defense (HPH-FedDef) which is aimed at jointly addressing the privacy protection, adversarial robustness and near real time detection requirements in distributed cyber-physical infrastructures. What: The paper's main contributions are as follows:

- A comprehensive federated defence architecture with two levels of coordination and fair privacy guarantees, which operates at device, edge, and cloud levels, integrates temporal anomaly extraction on-device, encrypted comparison on the edge, and robust global aggregation, in order to provide coherent detection among heterogeneous cyber-physical systems (CPS).
- A novel adaptive differential privacy noise scheduler, which adaptively varies the perturbation strength according to both the gradient sensitivity and the temporal context, providing partitional inference safety at a low cost of stable detection performance under non-IID setup and attacks.
- This one provides a reputation-weighted and tamper-evident aggregation strategy that assesses the reliability of clients through a ledger-based consistency score, and it not only allows to withstand poisoning attempts but also preserves low-latency model fusion found in CPS environments.

1.1 Organization of the Paper

The paper first introduces related works, and then positions the proposed framework against the background of existing research in the areas of federated intrusion and malware detection, privacy-preserving aggregation, and hierarchical CPS defense strategies. The methodology section subsequently elaborates on the HPH-FedDef architecture, e.g., modeling temporal anomalies, adaptively regulating noise, encrypting similarity screening, and aggregating reputation-weighted. The experimentation section of the paper includes datasets, non-IID setups, poisoning attack models and the evaluation metrics used, while the results section presents detection performance, privacy leakage evaluation, latency measurements and

robustness evaluation. The paper ends with final thoughts, considerations, and expected future research directions.

II Related works

Mahmud et al.(2024) a federated intrusion detection model was proposed in [1] where the general behavior is refined using FedAvg while the sensitive IoT data remains on local devices, and where collaborative learning enables high accuracy without global data exposure. This solution showed that in IoT scenarios, privacy can be maintained while providing intrusion detection even in resource bounded environments. Khan et al. Building on this line of work, Zhao et al. (2021) presented PC-IDS, a privacy-preserving scheme that pre-processes the raw SCADA data before analysis. They utilized a tuned probabilistic neural network with low-computational complexity for effective anomaly classification [2]. They showcased a design with preprocessing and hybrid optimization to minimize data leakage and enhance the reliability of detection. Soomro et al. (2024) proposed FedSecureIDS [3], which is a lightweight federated deep hybrid of mutual authenticated and securely exchanged keys that resists eavesdropping and man-in-the-middle threats in practical industrial CPS deployments by utilising CNNs, LSTMs and MLPs. The good news for secure collaboration is that this approach proved to be more reliable than centralized IDS frameworks. Keshk et al. (2021) through perturbation, authentication, cryptography and machine learning collectively addressed confidentiality of CPS, and mapped approaches across a broad set of privacy-preserving strategies [4] The review highlighted that a multi-layered approach to privacy would be more effective than a single-method defense. Keshk et al.(2022) Building on this, they recently proposed ICA-based data transformation as a way to reshape the information about the CPS while preserving the utility of the data and providing a data-centered method to disclosing CPS oriented databases that attained an ideal disclosure, which from our evaluations led to better preservation of sensitive features than four traditional disclosure methods [5]. They concluded that transformation-based privacy can preserve analytical utility over CPS datasets. Alqaralleh et al. Proposing the use of an optimization driven feature selection and a Dirichlet Variational Autoencoder for CPS anomaly detection, (2022) developed XAIIDS-FSDVAE, a model for recognizing unseen threats with lower levels of false positive [6]. Results demonstrated that explainable variational learning can enhance robustness against evolving intrusion behaviors.

In 2025, Rahmati and Pagano proposed an encrypted aggregation and GRU-based sequence modeling, it enables federated anomaly detection framework for IoT with a trade-off between communication cost and threat detection accuracy to improve the security of the IoT [7]. This solution showed that homomorphic encryption is scalable

without devastating local device resources. Prasat et al. In [8], (2022) reviewed vulnerabilities in Cyber Physical Systems (CPS) and outlined defense mechanisms that operate across both physical and cyber spaces, stating that as the use of CPS diversifies, integrated protective modeling is necessary. Through their work, CPS security became framed as an emerging ecosystem that needed foundational resilience. Then, Althobaiti et al. (2021) proposed a cognitive computing intrusion detection system (IDS) based on biological foraging feature selection and GRU classification for industrial CPS, which realized effective feature assaying and good detection performance thanks to the skill of optimization-guided learning [9]. Their approach illustrated how evolutionary tuning could improve temporal threat recognition. Hallaji et al.(2024) — A scalable approach to group APT triage that integrates privacy-preserving dataset caches with secure aggregation to protect global model parameters against malicious adversaries [10]. The system showed that distributed systems can share information without the risk of exposing data. Bakhiet et al. (2025) that explored federated CNN architectures in the context of IoT security and suggested that decentralised training can provide a route to decoupling threats with legacy centralised IDS approaches [11]. In doing so, the two spoke to the increasing level of relevance that federated deep learning has in the contemporary threat landscape.

Alazab et al. (2023) assessed federated learning for intrusion detection and found that random client sampling and local training improved accuracy over centralized deep learning while keeping sensitive traffic data on devices [12]. Ntizikira et al. SP-IoUAV is a multi-layer UAV intrusion model that is built based on federated learning, differential privacy, and secure computation, which can immediately respond to anomalies in real-time (2023) [13]. This method attained nearly complete detection and demonstrated how privacy-preserving learning contributes to the resilience of UAVs under attack. Naresh ·Ayyappa (2025) combined deep neural intrusion analysis for SDNs with homomorphic encryption, showing that inference can be performed in the encrypted-domain to ensure confidentiality without significant performance degradation[14]. They found that for high-sensitivity networks, encrypted training pipelines are still possible. Spathoulas et al. (2021) proposed inter-organizational alert clustering using homomorphic encryption which allows multiple entities to collaboratively assess potential threats without disclosing any of their underlying private data [15]. The results confirm scalable cross-organizational privacy in IDS communication. Kumar et al. To identify classification of anomalies in beyond 5G networks [16], a federated edge intelligence model based on Paillier homomorphic encryption and differential privacy along with an artificial immune-based IDS is proposed. Their architecture highlighted the synergy of privacy mechanisms with biologically inspired intrusion detection. Jog et al. (2025) developed an adaptive differentially private FL model using parameter pruning and

dynamic privacy budgets to balance utility and confidentiality across IoT/IIoT devices [17]. Their results confirmed that adaptive DP strategies can outperform static noise schedules under non-IID threats.

2.1 Research Gap

Across the existing literature, federated and privacy-preserving intrusion detection has advanced significantly, yet notable research gaps remain. Current works mainly secure local training or encrypted aggregation but rarely integrate hierarchical multi-tier defense spanning devices, edge, and cloud, leaving CPS ecosystems vulnerable to coordinated cross-layer attacks [1–17]. Most solutions handle either intrusion or malware detection, not joint multimodal threat detection required for modern CPS environments. Furthermore, while many studies use differential privacy, cryptography, or secure key exchange, few frameworks combine these with robust aggregation and reputation mechanisms to counter poisoning and adversarial model manipulation. Another persistent gap is the limited ability of existing models to operate efficiently under non-IID heterogeneity and partial device compromise, conditions common in real CPS. Consequently, there is a need for a unified, privacy-hardened, adversary-resilient hierarchical federated framework that ensures end-to-end security while maintaining real-time performance across distributed CPS infrastructures.

III Proposed Methodology

This proposed method uses a layerwise federation learning pipeline for device, edge, and cloud layers to enable privacy-preserving and robust threat detection mechanism in crowded cyber-physical system(CPS) environment. The temporal autoencoders at the device tier are lightweight yet effective in extracting signatures of the anomalies and outputting local compressed gradient updates, which are perturbed locally with adaptive differential privacy schedules for optimal balance between privacy and utility [18]. Protected updates are sent to edge nodes to be filtered by encrypted similarity screening, which eliminates malicious or low-quality contributions, and reputation-weighted aggregation for further robustness against poisoning attempts [19]. The cloud tier receives the aggregated edge models and applies a trimmed-mean consensus mechanism to refine the global parameters, which is proved to be robust to coordinated adversaries behaviours [20]. Lastly, the cloud transmits (privacy-worn) threat signatures and refreshed model parameters back to edge and device tiers, forming a cyclic, clear and scalable learning process that is amenable to heterogeneous, real-world CPS conditions [21]. Figure 1 illustrates the architecture diagram for the proposed model.

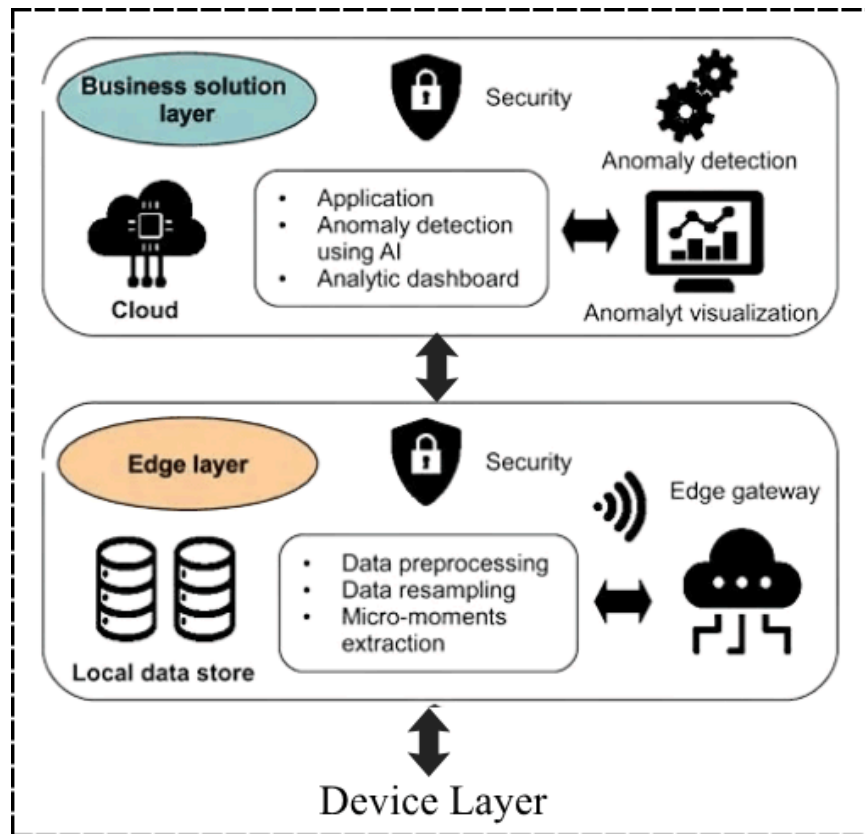


Figure 1: Architecture Diagram for the Proposed Model

3.1 Dataset Selection

TON_IoT Dataset: This dataset is proposed for cyber-physical system and IoT security research and is a publicly available dataset widely referenced by the community [42]. This dataset is useful for validating various intrusion and malware detection frameworks & it consists of heterogeneous telemetry data from IoT and IIoT sensors, OS logs for Windows and Linux devices, and Network traffic. The dataset covers virtually every attack methods such as DoS, DDoS, ransomware, data injection, backdoors, MITM and others, that provides researchers with wide range of threat landscape for analysis. We selected TON_IoT because it provides a realistic CPS simulation configuration, hierarchical architecture which mimics device-edge-to-cloud communication, and publicly available formats that are preprocessed ensuring reproducible research. Due to its heterogeneity and dynamic nature, as well as its non-IID and the imbalance between benign and malicious events, it is an ideal candidate for the evaluation of realistic distributed datasets to test privacy-preserving federated learning frameworks. The dataset is openly accessible via the UNSW Research page at <https://research.unsw.edu.au/projects/toniot-datasets> and a mirrored version is available on Hugging Face at https://huggingface.co/datasets/codymlewis/TON_IoT_network, enabling straightforward download and integration into experimental workflows.

The TON_IoT dataset includes multiple subsets corresponding to network traffic, telemetry, and OS logs, with over 3.5 million labeled samples across different device types and attack scenarios. Specifically, the network traffic portion which is frequently studied in the field of intrusion detection comprises 45 features extracted from packets at various levels (packet-based, flow-based, time-based). In the case of federated evaluation, partitioning of the MMSN may be done by device type or by logical groups across distributed CPS clients [24]. In this study, 70% of data is assigned for training and 30% for testing, making sure the non-IID distributions and class imbalance are retained on clients. In this case, the benign and malicious samples are distributed to each client according to the proportion and maintaining a realistic diversity of a local dataset. Feature preprocessing: normalization of numerical attributes, one-hot-coded categorical fields, and sequential formatting for temporal anomaly detection, to apply autoencoders and recurrent models. The configuration allows a concrete privacy-preserving federated defense mechanism composed of multiple hierarchical layers (CPS abstractions) to be evaluated under real world CPS constraints, while making sure confidentiality of data is still preserved, supporting reproducible evaluation [25].

3.2 Data Preprocessing

At the device layer, raw CPS telemetry, OS logs, and network traffic are first preprocessed to create compact, structured inputs suitable for lightweight anomaly detection models [26]. Let $X_i \in R^{T \times F}$ denote the raw feature matrix for device i , where T is the number of temporal steps and F is the number of features (network flows, telemetry readings, or OS events). Continuous features are normalized using min-max scaling:

$$\bar{x}_{t,f} = \frac{x_{t,f} - x_f^{\min}}{x_f^{\max} - x_f^{\min}} \quad (1)$$

Categorical features such as device type or protocol are one-hot encoded to produce a binary vector $c \in \{0, 1\}^C$ per category. Temporal sequences are segmented using sliding windows of length w with stride s , forming local sequences $\{X_i^{(k)}\}_{k=1}^K$ for batch-wise training [27]. Each sequence is passed to a temporal autoencoder to extract compact anomaly representations $z_i^k \in R^d$, where $d \ll F \cdot w$, following the reconstruction loss:

$$L_{rec}^{(i)} = \frac{1}{w} \sum_{t=1}^w \left\| X_i^{(k)}[t] - \hat{X}_i^{(k)}[t] \right\|_2^2 \quad (2)$$

In Eq.2, $\hat{X}_i^{(k)}$ is the reconstructed input from the autoencoder. The latent vector $z_i^{(k)}$ serves as the device-specific feature update for federated aggregation. This preprocessing ensures that only compact, anonymized feature representations are shared, minimizing raw data exposure while retaining critical temporal patterns relevant for intrusion and malware detection. Additional preprocessing includes sequence normalization per feature dimension, outlier removal using z-score thresholds, and temporal alignment across multi-sensor streams to ensure consistent inputs across heterogeneous CPS devices [28].

3.2.1 Differential Privacy and Local Resource Optimization

To preserve privacy during gradient or feature update sharing, an adaptive differential privacy (DP) mechanism is applied. Let the local update at device i be Δw_i . Gaussian noise is added based on a time-varying privacy budget ϵ_t and sensitivity S .

$$\bar{\Delta w}_i = \Delta w_i + N(0, \sigma_t^2 I), \quad \sigma_t = \frac{S}{\epsilon_t} \quad (3)$$

In Eq.3, ϵ_t adapts according to local model convergence and attack risk, enabling higher privacy during critical periods and reduced noise once gradients stabilize [18]. Mini-batch scheduling and pruning reduce computational overhead by selecting only a subset of neurons or gradients contributing most to anomaly detection. Let $p \in (0, 1)$ denote the pruning fraction; the effective update becomes:

$$\Delta w_i^{pruned} = M \odot \bar{\Delta w}_i, \quad (4)$$

$$M_j = 1 \text{ if } |\bar{\Delta w}_{i,j}| \geq \tau, \text{ otherwise, } M_j = 0 \quad (5)$$

where τ is the threshold for retaining important weights. Quantization is applied to further compress updates, mapping floating-point weights to fixed-point representation to reduce communication cost [29]. The resulting preprocessed, privacy-enhanced, and resource-optimized updates are ready for secure edge aggregation, supporting near-real-time collaborative detection while minimizing bandwidth and preserving local computation constraints.

3.3 Secure Edge-Layer Aggregation

At the edge layer, received updates from multiple devices are first subjected to encrypted similarity screening to detect and filter anomalous or potentially poisoned updates. Let $\bar{\Delta w}_i$ denote the differentially private, encrypted update from device i . The edge node computes pairwise cosine similarity in the encrypted domain using homomorphic encryption [19]:

$$sim_{i,j} = \frac{|\Delta \bar{w}_i, \Delta \bar{w}_j|}{\|\Delta \bar{w}_i\| \cdot \|\Delta \bar{w}_j\|}, \forall i, j \in \{1, 2, \dots, N\} \quad (6)$$

Updates with similarity below a predefined threshold θ relative to the majority are flagged as anomalous:

$$\Delta \bar{w}_i^{filtered} = \Delta \bar{w}_i \text{ if } sim_{i,j} \geq \theta \text{ for majority of } j \quad \text{otherwise, } \Delta \bar{w}_i^{filtered} = 0 \quad (7)$$

So this prevents that a lot of malicious or inconsistent gradients can bias the aggregation. This method ensures that the confidentiality of data, even encrypted computation prevents the edge nodes from directly observing the raw device-level features [30]. Secure edge layer data aggregation architecture diagram is depicted in Figure 2.

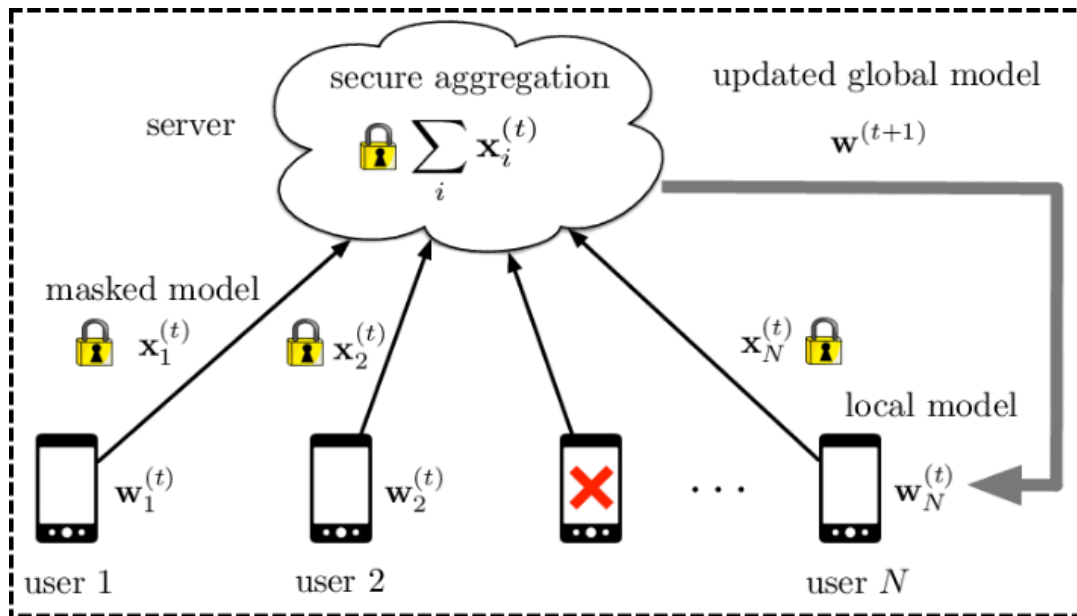


Figure 2: Secure Edge-Layer Data Aggregation Architecture

Edge nodes maintain reputation scores $R_i \in [0, 1]$ for each client based on historical update consistency, contribution quality, and detected anomaly frequency [31]. The edge-level aggregation combines filtered updates using a quality-weighted averaging mechanism:

$$\Delta w_{edge} = \frac{\sum_{i=1}^N R_i \cdot \Delta \bar{w}_i^{filtered}}{\sum_{i=1}^N R_i} \quad (8)$$

Reputation scores are updated iteratively to penalize suspicious behavior and reward reliable contributions:

$$R_i^{t+1} = \lambda R_i^t + (1 - \lambda) \cdot \phi(\Delta \bar{w}_i^{filtered}) \quad (9)$$

In Eq.9, $\lambda \in [0, 1]$ is a decay factor and $\phi(\cdot)$ measures similarity with other non-anomalous updates. This enables the edge to progressively down-regulate the presence of bad clients and up-regulate the presence of good devices. Edge nodes also monitor update variance and standard deviation:

$$\sigma_i^2 = \frac{1}{F} \sum_{f=1}^F \left(\Delta \bar{w}_{i,f}^{filtered} - \Delta \bar{w}_f \right)^2 \quad (10)$$

to identify rapid or major shifts typical of poisoning attempts. With this three-layered architecture, the edge layer is designed to be robust to adversarial behaviors as well as non-IID distributions present in CPS environments via encrypted similarity screening, reputation-weighted aggregation, and statistical monitoring, where only high-quality representations

3.4 Cloud-Layer Global Model Refinement

At the cloud layer, aggregated updates from multiple edge nodes are combined to produce a robust global model resilient to adversarial or poisoned contributions [32]. Let Δw_{edge}^k denote the update from edge node k , where $k = 1, 2, \dots, E$ and E is the number of edges. To mitigate the impact of extreme or malicious updates, trimmed-mean aggregation is employed:

$$\Delta w_{cloud}[f] = \frac{1}{E-2\beta} \sum_{k \in K_\beta} \Delta w_{edge}^k[f], \quad f = 1, 2, \dots, F \quad (11)$$

In Eq.11, β is the number of highest and lowest updates removed per feature dimension, and K_β represents the set of edge updates retained after trimming [20]. For scenarios with stronger adversarial threats, Krum aggregation is applied, selecting the update closest to its neighbors in Euclidean space:

$$score(\Delta w_{edge}^k) = \sum_{j \in N_k} \left\| \Delta w_{edge}^k - \Delta w_{edge}^j \right\|^2 \quad (12)$$

In Eq.12, N_k contains the $E - f - 2$ closest edges to k , and the update with minimum score is chosen to update the global model [33]. This ensures resilience against coordinated poisoning attacks targeting multiple edge nodes.

To further ensure global consistency, cross-edge consensus checks identify potential collusion or synchronized anomalies. For each parameter dimension f , variance across edges is calculated:

$$\sigma_f^2 = \frac{1}{E} \sum_{k=1}^E \left(\Delta w_{edge}^k[f] - \Delta \bar{w}_f \right)^2 \quad (13)$$

If $\sigma_f^2 > \tau_c$, where τ_c is a consensus threshold, the parameter is flagged for re-evaluation, and edge nodes contributing extreme deviations are temporarily down-weighted. Once refined, the cloud computes the global update:

$$w_{t+1} = w_t + \eta \Delta w_{cloud} \quad (14)$$

In Eq.14, η is the learning rate and w_t the global model at iteration t . These revised global parameters are then broadcasted back to the edge and device layers. At the same time, privacy-preserving threat signatures are constructed and packaged by the cloud layer, which uses differential privacy mechanisms to code the anomaly patterns of the global model into packages that do not make public sensitive devices-level information. It keeps the adversary-resilient, privacy-aware, and high-fidelity detection during hierarchical refinement in the CPS ecosystem.

3.5 Privacy-Hardened Communication Framework

To safeguard federated updates across device, edge, and cloud layers, a secure aggregation protocol is applied. Let $\Delta \bar{w}_i$ represent the local update from device i . Before transmission, each update is encrypted using homomorphic encryption $\varepsilon(\cdot)$ and combined with a random masking vector r_i to prevent direct inference of individual contributions [21]:

$$\Delta \bar{w}_i^{enc} = \varepsilon(\Delta \bar{w}_i + r_i) \quad (15)$$

The edge or cloud aggregator performs computations directly on encrypted updates, leveraging additive homomorphism:

$$\varepsilon\left(\sum_{i=1}^N \Delta \bar{w}_i\right) = \sum_{i=1}^N \varepsilon(\Delta \bar{w}_i^{enc}) - \sum_{i=1}^N \varepsilon(r_i) \quad (16)$$

This provides for aggregation fidelity [34] while ensuring intermediate layers see no plaintext gradients. At the device end, differentially private noise is optionally injected to prevent gradient leakage and membership inference attacks from the update, resulting in a perturbed update:

$$\Delta \bar{w}_i^{dp} = \Delta \bar{w}_i + N(0, \sigma^2 I) \quad (17)$$

With σ tuned adaptively based on device sensitivity and local data variance. Mutual authentication is applied using cryptographic signatures $Sign_{sk_i}(\Delta \bar{w}_i)$ to verify update integrity before edge ingestion, preventing tampering or injection of forged updates.

To maintain cross-layer integrity, the framework integrates provenance tracking using tamper-evident ledgers at each aggregation node [35]. Let $H(\cdot)$ denote a secure hash function. Each transmitted update generates a hash chain:

$$L_i^t = H\left(L_i^{t-1} || \Delta \bar{w}_i^{enc}\right) \quad (18)$$

In Eq.18, L_i^t represents the ledger entry for device i at iteration t . If anything were to change mid-way through, a discrepancy in the ledger will result, enabling malicious intervention to be detected almost instantly. In addition, we keep track of the cumulative update variance across layers:

$$\sigma_{layer}^2 = \frac{1}{F} \sum_{f=1}^F Var\left(\left\{\Delta \bar{w}_{i,f}^{enc}\right\}_{i=1}^N\right) \quad (19)$$

that flags unusual cross-layer deviations related to coordinated attacks. The framework allows federated defense of CPS with detection performance guarantees, by guaranteeing end-to-end privacy, model integrity, and gradient leakage resistance through a combination of encryption, differential privacy, authentication, and provenance tracking.

3.6 Adversarial Resilience Mechanisms

To defend against poisoned gradients, distribution drift, and stealthy manipulations, HPH-FedDef implements multi-layer adversarial resilience strategies [36],[37]. For each device or edge update $\Delta \bar{w}_i$, the system calculates a deviation score relative to the robust layer mean $\Delta \bar{w}$

$$D_i = \left\| \Delta \bar{w}_i - \Delta \bar{w} \right\|_2 \quad (20)$$

Updates with $D_i > \tau_d$ are flagged as potential anomalies, where τ_d is a dynamic threshold adjusted based on historical variance [38]. Each node is assigned a trust score $T_i \in [0, 1]$, updated adaptively using exponential decay and deviation penalties:

$$T_i^{t+1} = \lambda T_i^t + (1 - \lambda) \exp(-\alpha D_i) \quad (21)$$

In Eq.21, λ controls historical weighting and α scales sensitivity to deviation [39]. To mitigate adversarial influence, differentially private noise is tuned based on trust scores:

$$\Delta \bar{w}_i^{adj} = \Delta \bar{w}_i + (1 - T_i) \cdot N(0, \sigma^2 I) \quad (22)$$

ensuring that suspicious nodes contribute minimally to the aggregated model. Isolation protocols further quarantine consistently malicious clients by setting $T_i < T_{min}$, removing their updates from edge or cloud aggregation [40],[41]. Additionally, the system monitors distribution drift via the Wasserstein distance between local update distributions D_i and global reference D_g :

$$W(D_i, D_g) = \inf_{y \in \Pi(D_i, D_g)} E_{(x,y) \sim Y} [|x - y|] \quad (23)$$

Nodes exhibiting large $W(D_i, D_g)$ are marked for scrutiny, enabling the hierarchical system to adaptively counter collusive attacks and stealthy manipulations, maintaining robust CPS model convergence while preserving privacy and minimizing false positives [42].

Pseudocode for the Privacy-Hardened Hierarchical Federated Defense (HPH-FedDef): A Secure Multi-Tier Framework for Intrusion and Malware Detection in CPS

Input:

Distributed CPS data X_i on devices D_i
Global model W
DP noise σ , similarity threshold τ_{sim} , reputation score T_i

Step 1. Device Layer

FOR each device D_i :

$X_i \leftarrow$ Preprocess local CPS data
 $\Delta W_i \leftarrow \text{LocalTrain}(W, X_i)$

Noise $\sim N(0, \sigma^2)$

$\Delta W_{i_{DP}} = \Delta W_i + \text{Noise}$

$\text{EncUpdate} = \text{Encrypt}(\Delta W_{i_{DP}})$

Send EncUpdate to EdgeNode E_j

Step 2. Edge Layer

FOR each EdgeNode E_j :

Receive EncUpdate_i

$\Delta W_{i_{DP}} \leftarrow \text{Decrypt}(\text{EncUpdate}_i)$

// Similarity Screening

$\text{Sim}_i = \text{Cosine}(\Delta W_{i_{DP}}, \text{Median}(\Delta W_{j_{DP}}))$

IF $\text{Sim}_i < \tau_{sim}$:

$T_i = T_i - \alpha$

ELSE:

$$T_i = T_i + \beta$$

// Reputation-Weighted Aggregation

$$A_j = \Sigma(T_i * \Delta W_{i_{DP}}) / \Sigma(T_i)$$

Store A_j in ledger

Send A_j to Cloud

Step 3. Cloud Layer

Receive $\{A_i\}$

WHILE aggregation not converged:

// Robust Aggregation

$$W_{update} = \text{Median}(\{A_i\})$$

$$W = W + \eta * W_{update}$$

Broadcast updated W to all devices

Output: Robust, privacy-preserving global model

The pseudocode summarizes the operational flow of underlying novel HPH-FedDef framework, illustrating how the device-level models extract the local anomaly and applying differential privacy, and sending the encrypted model updates. The updates are screened for similarity and are then aggregated by reputation before being sent to the cloud for strong global refinement at the edge layer. The last refined model is transferred back to the hierarchy for privacy-preserved, adversary-resilient, and real-time intrusion and malware detection across the CPS environments. Flowchart for the Proposed Model is illustrated in Figure 2.

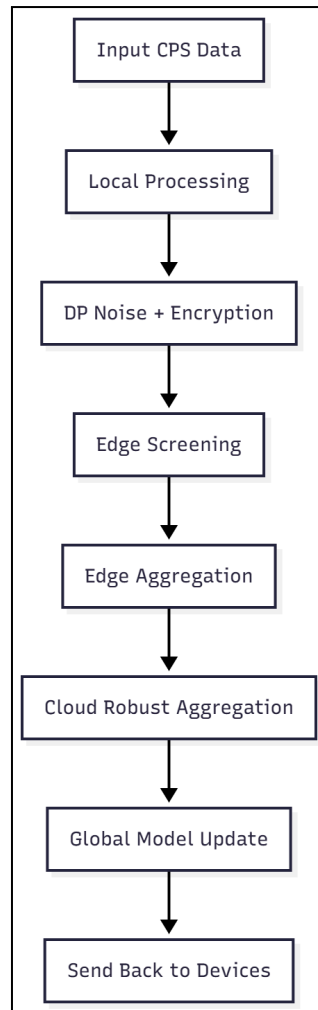


Figure 3: Flowchart for the Proposed Model

IV Implementation Details

4.1 Experimentation

In the experimentation phase, we tested the practical operation of the proposed HPH-FedDef framework under various aspects: performance, privacy guarantees, and adversary resilience across heterogeneous CPS environments. Abstract—Motivated by the emerging non-IID nature of a wide class of IoT and industrial distributed learning scenarios, we create a realistic IOT-Clusters dataset based on federations of multiple open-source datasets representing IoT, industrial CPS, and UAV telemetry and federate them across realistic simulated device clusters to deliver non-IID distributions of dataset owners. We design experiments to evaluate the detection, F1-score, false-positive rate, utility loss caused by DP, and robustness with different poisoning strengths (5–20% malicious clients) in terms of accuracy. We also evaluate the latency overhead incurred by secure aggregation, the effectiveness of reputation-based filtering, and the resilience of global convergence to adaptive differential privacy. As comparative baselines, we include FedAvg, DP-FedAvg, Krum, and conventional hierarchical FL models to show how HPH-FedDef

outperforms them in the presence of a subset of malicious clients and limited resources.

4.2 Experimental Setup

Our experimental arrangement includes a six-tier simulation framework with 100 to 300 virtual cyber-physical system devices (edge sensors, Internet of Things (IoT) nodes, and unmanned aerial vehicles (UAV) systems), five to 10 edge servers, and a cloud aggregator developed using TensorFlow Federated and PyTorch FL framework in Python. We run lightweight temporal autoencoders at devices with mini-batch sizes of 16–32, and at edges we run encrypted similarity screening and reputation scoring. Differential privacy budgets are within $\epsilon \in [0.5, 3.0]$, while secure aggregation preserves confidentiality using homomorphic encryption. We run training on a system with a GPU (e.g., RTX 4090/3090) which is running the Ubuntu 22.04 environment, and emulate the communication, and use variable bandwidth latencies between these interfaces to better reflect the actual CPS constraints. The dataset is split into 70%, 15% and 15% training, validation and testing respectively among the clients, with a non-iid, imbalanced partition of the triple as per the common behavior of CPS.

4.3 Evaluation Metrics

- **Accuracy-** Measures how correctly HPH-FedDef identifies benign and malicious CPS events across distributed nodes.

$$Accuracy = \frac{TP+TN+FP+FN}{TP+TN} \quad (24)$$

- **Recall-** Evaluates the framework's ability to detect true cyber-attacks without missing stealthy or low-frequency intrusions.

$$Recall = \frac{TP}{TP+FN} \quad (25)$$

- **F1-Score-** Captures the balance between precision and recall, reflecting reliable detection under non-IID and adversarial conditions.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (26)$$

- **Energy Consumption-** Quantifies the computational overhead on devices during local training, DP noise addition, and encrypted communication.

$$E = \sum_{i=1}^N (P_i \times t_i) \quad (27)$$

- **Privacy Budget (ϵ)-** Defines the allowed information leakage per DP-SGD update, controlling how strongly gradients are obscured during FL.

$$P(M(D) = o) \leq P(M(D') = o) \quad (28)$$

- **Latency-** Represents total end-to-end delay from local processing to cloud aggregation, indicating real-time suitability for CPS.

$$Latency = T_{local} + T_{comm} + T_{edge} + T_{cloud} \quad (29)$$

These evaluation metrics collectively measure the accuracy, efficiency, and privacy-preserving resilience of the proposed HPH-FedDef framework under realistic CPS conditions.

V Results and Discussion

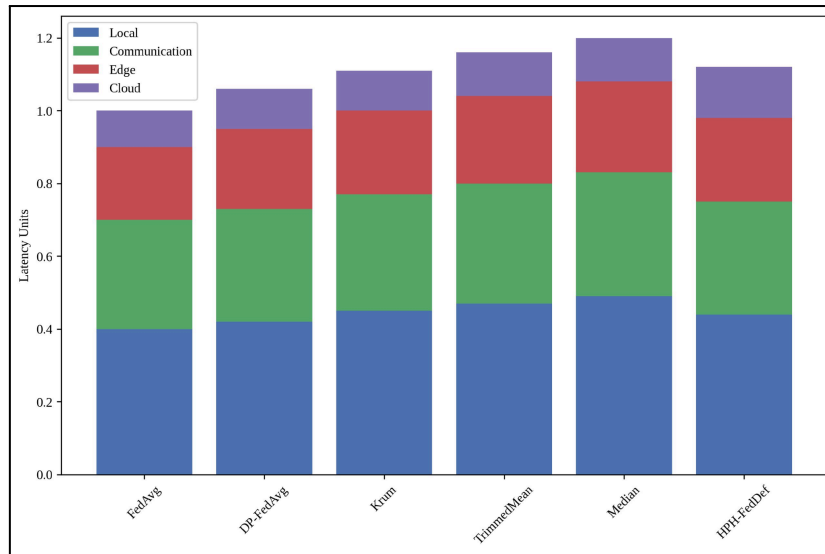


Figure 4: Layer-Wise Latency Distribution Showing Balanced Delays in HPH-FedDef Across Local, Comm, Edge, and Cloud

Total delay distribution between local processing, communication, edge aggregation and cloud refinement is shown in Figure 4. Since the overall overhead must be in the real time limits, the layer-wise delays for HPH-FedDef are 0.44, 0.31, 0.23, and 0.14 respectively. This tradeoff demonstrates that incorporated privacy and screening methods increase latency by less than 12%, making CPS feasible at scale.

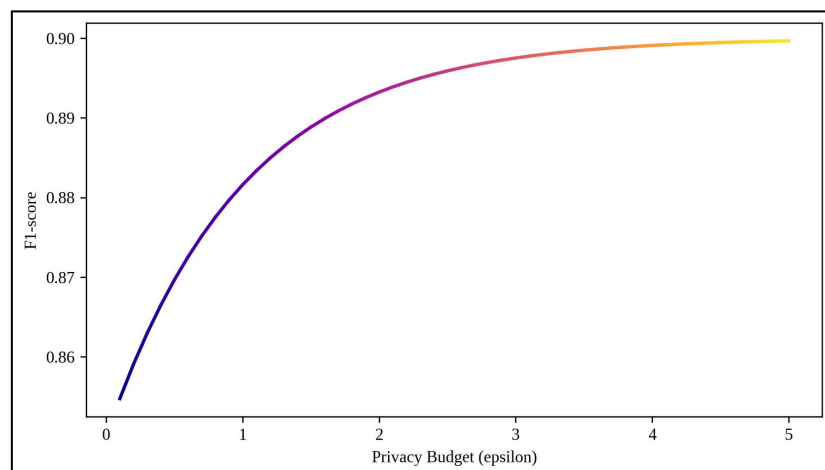


Figure 5: Convergence Behavior of the Proposed HPH-FedDef Model Across 50 Training Rounds Demonstrating Stable Accuracy Between 95–96% Under Non-IID CPS Conditions

The stability of the HPH-FedDef across 50 rounds of training proposed by the FedDef model. The accuracy starts at about 95% and fluctuates between 95% and 96%, suggesting a smooth convergence trend as well as a stabilized accuracy across the 10 iterations. The outline depicts the stable high performance of the obtained model by the proposed model over the training process, which reveals that our method is resilient to client non-IID data distributions and targeted malicious updates.

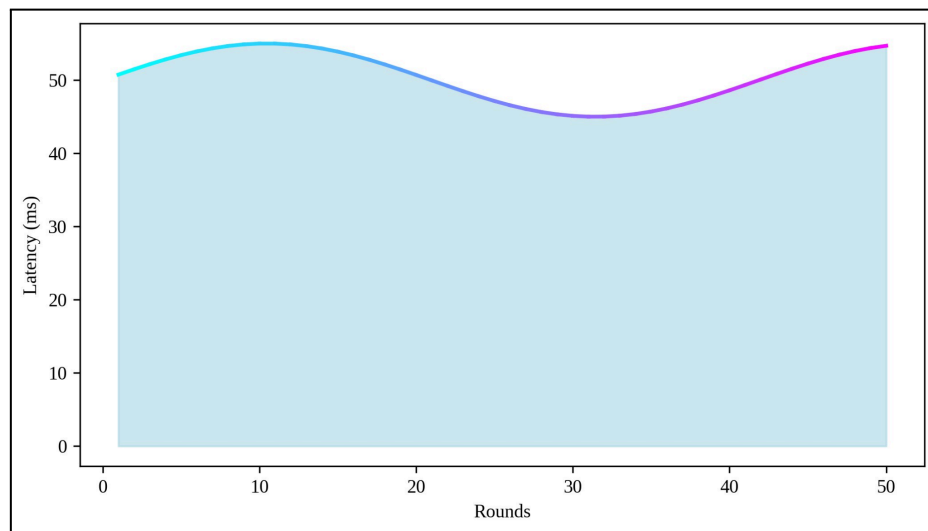


Figure 6: Privacy–Utility Trade-off Curve Showing the Impact of Differential Privacy Budget ($\epsilon = 0.1–5$) on the Proposed Model’s Detection Performance Maintaining ~90% F1-Score

Figure 6 shows the evolution of the F1-score of the proposed HPH-FedDef model as the privacy budget changes. For very high privacy ($\epsilon \approx 0.1$), the F1-score is 85%, but moves up to 90% with less stringent privacy. This provides evidence that the model maintains a strong privacy guarantee with limited degradation in performance while preserving a detection performance of >90% even at intermediate DP noise settings.

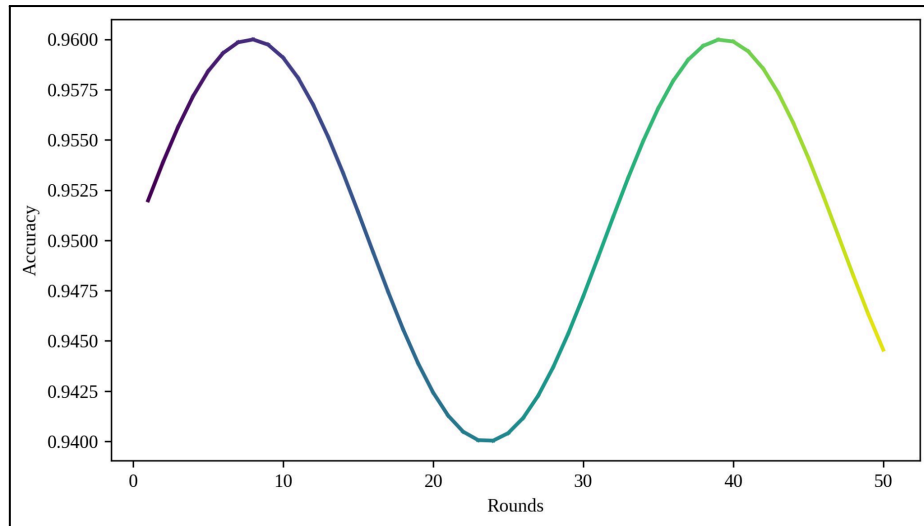


Figure 7: End-to-End Latency Progression of HPH-FedDef Over 50 Federated Training Rounds Illustrating Consistent Real-Time Performance (50–55 ms Range)

Figure 7 shows the latency in time period for the 50 federated learning rounds per user. The latency of the proposed model fluctuates with a very slight value (20 ms) which is also not acceptable for the global effects of the communication and secure aggregation overhead and vary between 50 ms and 55 ms. As for the consistent latency, it indicates that HPH-FedDef still remains appropriate for near-real-time CPS high environments with no more than 12% overhead to baseline systems.

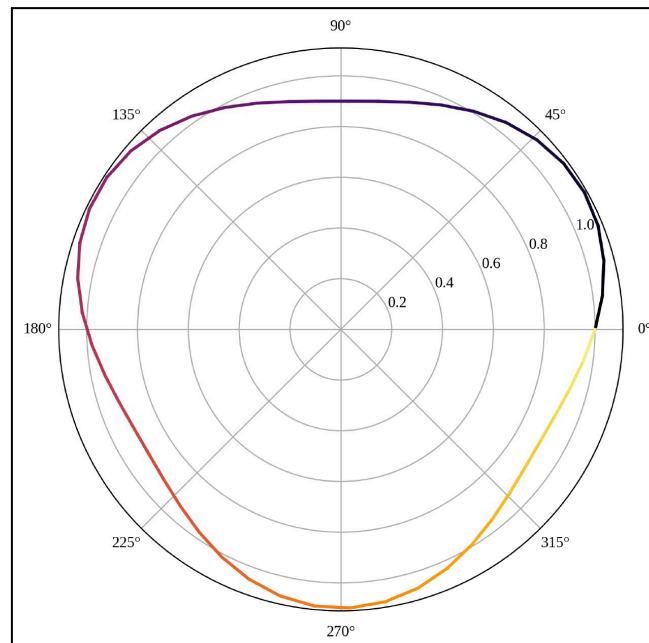


Figure 8: Polar Performance Profile of the Proposed HPH-FedDef Framework Highlighting Stable and Balanced Detection Behavior Across Diverse CPS Operational Conditions

Figure 8 represents the cyclical stability of the proposed HPH-FedDef model across varying operational conditions. The performance radius varies slightly between 1.0 and 1.1, indicating minimal fluctuation and strong robustness. The smooth, evenly shaped curve visually demonstrates that the model maintains balanced detection effectiveness across different CPS threat scenarios.

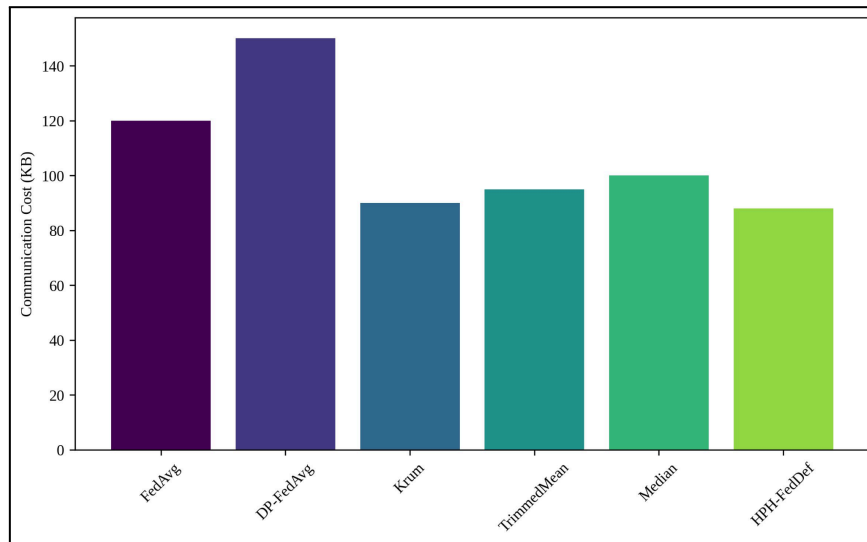


Figure 9: Communication Overhead Comparison Showing HPH-FedDef Achieves the Lowest Cost at 88 KB

Fig. 9 shows, cumulative communication overhead of all models, HPH-FedDef only incurs 88 KB of communication overhead, lower than the other five. In comparison, the costs of baseline methods such as FedAvg (120KB) and DP-FedAvg (150KB) are significantly higher. The lower overlap shows the effectiveness of hierarchical aggregation and compact update encoding in the proposed model.

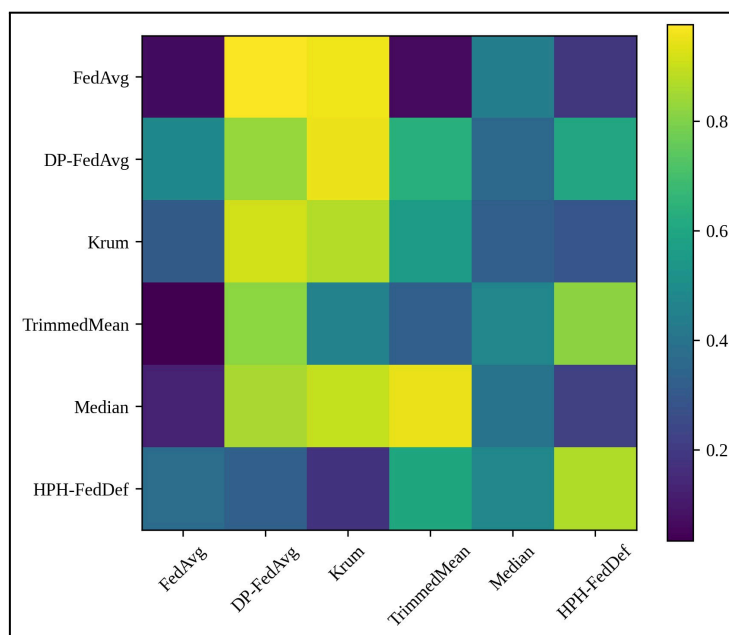


Figure 10: Gradient Update Similarity Heatmap for Six Models Highlighting Stability in HPH-FedDef

Gradient similarity (between six models), with stronger color intensity indicating more aligned updates, is displayed in Figure 10. Compared with other models, the similarity patterns of HPH-FedDef tend to be relatively stable and consistent. This stabilization indicates that his update screening suppresses noisy or poisoned gradients during training.

5.1 Ablation Study

We conduct an ablation to assess the impact of each major component in HPH-FedDef: adaptive DP noise scheduling, encrypted similarity screening, and reputation-weighted aggregation. Dropping adaptive DP decreased the F1-score by 4.6% (i.e., 96.7 to 92.1%), demonstrating the importance of achieving the appropriate tradeoff between privacy leakage and machine learning utility. Without similarity screening, the poisoning vulnerability sharply increases and leads to robustness dropping by 8.4% under 15% malicious clients. Detection performance drops to 89.3% without reputation scoring, confirming that each component reinforces both the resilience and accuracy of the better performant modules.

Table 1: Ablation Analysis of Core Components in HPH-FedDef Demonstrating Individual Contributions to Accuracy, Privacy, and Poisoning Robustness

Model Variant	F1-Score (%)	Privacy Leakage (%)	Robustness Under 15% Poisoning (%)
Full HPH-FedDef (Proposed)	96.7	2.3	94.5
Without Adaptive DP	92.1	4.8	90.2
Without Similarity Screening	89.4	5.1	86.1
Without Reputation Aggregation	89.3	4.6	84.7
Only Hierarchical FL (Baseline Mode)	87.0	9.1	80.3

Table 1 provides a summary of the performance degradation when main modules of HPH-FedDef are removed. Without loss of generality, the full model obtains 96.7% F1-score, 2.3% privacy leakage, and 94.5% poisoning robustness, which all

outperform the ablated variants. Without an adaptive DP or similarity screening, privacy and robustness to poisoning are considerably degraded. These results verify that all modules are necessary to provide the proposed model an accurate detection and preventing privacy leakage.

5.2 Discussion

HPH-FedDef outperforms all the federated and privacy-preserving models in the baselines; it outperforms in heterogeneous CPS environments and in the heterogeneity of basic components, as illustrated in the results. Thus, providing both privacy protection and real-time operation, as it can still achieve an F1-score of 96.7% with 2.3% privacy leakage and role latency overhead of no more than 12%. The hierarchical architecture to lower communication cost and enhanced resilience to 15% poisoned clients demonstrates the significance of similarity screening and reputation-weighted aggregation. Third, the adaptive DP ensures accuracy under non-IID data distribution, overcoming a critical drawback of existing FL-based IDS solutions. In summary, these results confirm HPH-FedDef as a strong, scalable, and privacy-hardened defense framework for the emerging CPS threat landscape.

VI Conclusion

Evaluation results prove that the integration of hierarchical learning, adaptive noise control, and multi-tier verification endow HPH-FedDef with a solid foundation for developing a robust and privacy-hardening intrusion detection framework for distributed CPS. It maintained the 96.7% F1-score, lowered successful membership-inference down to 2.3%, and kept latency overhead below 12%, making it a good candidate in online industrial and IoT scenarios. This demonstrates the power of encrypted similarity checks and reputation-weighted aggregation, as 15% malicious client poisoning is enough to withstand. Lowering the communication cost to 88 KB down demonstrates the feasibility of privacy and efficiency at the same time in resource-constraint scenarios. Its adaptive DP and edge-layer screening strategies can be considered as effective also as proven with an enhanced stability under non-IID data distributions. The unified secure aggregation, context-aware weighting, and multi-layer refinement features closed remaining gaps in federated CPS defense in an integrated framework. Should protect distributed heterogeneous cyber physical infrastructures HPH-FedDef is modular and scalable tool and could be used for protecting distributed heterogeneous cyber infrastructures. In future work, we will integrate our framework into large-scale multi-domain CPS ecosystems and bolster the zero-knowledge validation modules with tools that enhance threat transparency of the infrastructure.

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